

Data Wrangling

↳ it is the process of cleaning, organizing and preparing raw data for analysis.

What it includes :-

- Removing incorrect data
- filling missing values.
- converting data into proper format.

Importance :-

- Make data clean and usable
- Improve accuracy of results
- helps in better decision making.

② String Manipulation

↳ means working with and modifying text data.

Common Operation :-

text = "hello"

text.upper() → HELLO

text.strip() → "hello"

text.replace("hello", "hi") → "hi"

text.split() → ['hello']

importance

- Used to clean and format text.
- important in NLP, chatbots.
- helps in searching and filtering text data.

Handling large Data sets Efficiently

Techniques :-

- use efficient libraries (pandas)
- load data in small chunks instead of full data.
- Remove unnecessary columns.
- Avoid loops / built-in-functions.

Importance :-

- Saves money
- Increase processing speed
- prevent system crash.
- Improves scalability

④ Data Transformation Techniques

↳ data transformation means changing data into a suitable format for analysis or modeling.

Techniques :-

- Normalization → scale values b/w 0 and 1
- Standardization → mean = 0, variance = 1
- Encoding
- Aggregation → combine data
- Smoothing → remove noise.
- Discretization → group continuous data.

Importance :-

- improve model performance

- make data more understandable
- reduce errors and noise
- make algorithm work efficiently.